Layered representation diagnostics for ultra-short metagenomic reads

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Abstract

Short-read metagenomic next-generation sequencing (mNGS) pipelines often process trimmed, ambiguous or locally perturbed reads. Their useful sequence evidence may be compressed before classifiers or database lookups are applied. In this setting, accuracy-led evaluation is incomplete because downstream performance alone cannot show whether losses of identity, biochemical or positional information arise in the representation layer. Here we introduce a representation-diagnostics framework for controlled short-read (69-150 bp) mNGS settings. The framework dissects representations into four quantifiable dimensions: exact identity retention, biochemical perturbation stability, positional readability and local mutation sensitivity. Our compact representation, CK4P-MSP, combines reverse-complement canonical 4-mer identity with biochemical summaries and multi-scale property pooling. It is evaluated against full-position matrices as a positional upper bound and canonical spaced-property (CSP) as a boundary comparator. Across controlled WGS-derived perturbations, ART simulations, CAMI_TOY_low readout probes, a CAMI II marine anonymous-read stability probe and local mutation sensitivity tasks, compact property-aware representations showed lower standardized perturbation drift while retaining low-dimensional representation readability. Full-position matrices further showed that fine-grained positional information can add readout value at substantially higher dimensional cost, whereas spaced-seed priors transferred to dense read-level descriptors only in scenario-specific contexts. These results support a bounded architectural implication: exact k-mer, alignment and curated database evidence should remain the identity backbone, while compact biochemical and coarse position-aware summaries can provide block-decomposable coordinates for representation-level perturbation auditing.

Keywords metagenomics, short reads, representation diagnostics, k-mer features, biochemical sequence encoding

Metagenomic next-generation sequencing can interrogate complex or unexpected microbial mixtures without a fixed target panel, and it has become an important route for infectious-disease investigation when targeted assays are insufficient [1, 2, 3]. Yet the reads entering downstream analysis are often short, trimmed, ambiguous and locally corrupted. Adapter removal, quality trimming and low-biomass contamination can all change the effective sequence evidence available to a pipeline before classification begins [4, 5, 6]. In controlled short-read (69-150 bp) mNGS settings, these physical constraints do not mean that classification is impossible. They can still reduce the identity margin relative to longer reads and make it harder to see which categories of information remain available. A read representation may already have compressed local identity, biochemical regularity or positional information before a downstream benchmark reports success or failure. For this reason, short-read representation is not merely a preprocessing step. It defines which categories of information remain available to any later classifier, alignment routine or database query. In this manuscript, the 69 and 75 bp settings are used only as representative lower-read-length conditions; the underlying clinical data are not publicly available and are not used as experimental input.

Current metagenomic workflows are strongest when they preserve exact or near-exact identity evidence. Canonical k-mer, minimizer, alignment and database-backed systems such as *Kraken*-family tools, *Centrifuge*, *CLARK*, *Kaiju* and related indexing frameworks have established the practical value of explicit sequence matching for taxonomic assignment and downstream biological interpretation [7, 8, 9, 10, 11]. Community benchmarks further show that performance depends on database coverage, sample composition, novelty and evaluation context, not only on the classifier name [12, 13]. These methods should therefore be acknowledged rather than displaced: exact identity evidence remains the backbone for tasks such as species discrimination, allele-sensitive interpretation and resistance-gene calling. Spaced-seed indexing further extends this line by tuning match patterns for sensitivity under mutation, but its goal is still matching accuracy rather than layered diagnostic analysis [14]. The value of additional audit coordinates can become more visible in 69-75 bp lower-read-length settings, where exact identity evidence has less redundancy than at 150 bp. At the same time, these systems are designed to answer a different question from the one asked here. They determine what a read matches, but they do not directly diagnose which kinds of information remain visible in the representation itself when reads are short, noisy or partially degraded.

Vectorized k-mer representations provide the first broad alternative to explicit matching. Here the central operation is to collapse each read or fragment into a composition profile, such as count, frequency, relative-frequency, TF-IDF, or background-adjusted features. PlasFlow and the BMC Bioinformatics 2018 bacteria classifier show that this compact route can be highly competitive when the task is composition-driven rather than order-driven [15, 16]. Resource-saving k-mer classification extends the same logic by showing that low-dimensional k-mer distributions plus classical machine learning can remain competitive even when deep models are not the only option [17]. However, these vectorized k-mer methods rely solely on composition statistics and do not explicitly layer biochemical stability or positional information alongside identity evidence. That distinction becomes more consequential in 69-75 bp reads, where local composition survives only in compressed form. These methods are not weak baselines; they are a distinct representation family whose success depends on how much local composition survives the short-read regime.

Hashing and sketching form a separate compression family rather than a lower-rank version of SVD. MinHash, LSH, histosketch, and related sketching strategies map k-mer sets or profiles into fixed-length signatures for rapid similarity search, large-scale database indexing, or ultra-lightweight compression [18, 19]. In the local literature map, Kssd and the HULK/histosketch line fill this role most clearly: they compress the representation in service of indexing or neighborhood search, not in service of preserving linear variance structure. That distinction matters here because our vector route is not asking for the cheapest signature possible; it is asking whether a sparse composition vector retains linearly accessible signal once the short-read regime and the downstream probe are fixed. Sketching papers therefore belong in the related work, but as a neighboring compression paradigm, not as a substitute for vectorized k-mer analysis.

Biochemical and signal-based encodings add a second alternative representation family. Chaos-game and genomic-signal formulations, together with numerical mappings such as Voss, tetrahedron, and electron-ion interaction potential (EIIP)-style encodings, expose structure that is not reducible to literal token identity alone [20, 21, 22, 23, 24, 25, 26]. These encodings matter because they keep the question on the representation itself: what survives after a read is translated into a signal, not only what survives after it is matched. In lower-read-length settings, such encodings are especially relevant because they can expose stability patterns when exact identity evidence has a smaller margin. However, existing biochemical encodings are usually used as standalone features rather than layered alongside exact identity evidence, and their perturbation-stabilizing value under controlled short-read conditions remains largely unquantified.

Sequence-token deep learning classifiers make the neighboring problem space especially relevant. DeepMicrobes introduced deep learning for metagenomic taxonomic classification, using canonical k-mer tokenization, embedding, BiLSTM and attention for read-level assignment [27]. MetaTransformer extended the same general family with self-attention and alternative tokenization or embedding schemes, including character-level, BPE, k-mer and hash/LSH variants [28]. BERTax showed that a small-k Transformer can support hierarchical taxonomy if the pretraining and output heads are aligned to the task [29]. BarcodeBERT then demonstrated that barcode-scale biodiversity tasks can benefit from domain-specific pretraining and masking choices, but its setting is still a barcode-specific classification problem rather than shotgun short-read mNGS [30]. Large pre-trained DNA language models further extend this representation family as general-purpose encoders [31, 32, 33, 34, 35]. These learned embeddings can be strong, and in some downstream probes they may outperform compact hand-defined representations. The question asked here is different. We do not test whether CK4P-MSP is a stronger predictive encoder than a Transformer. We ask whether a low-dimensional, training-free and block-decomposable representation can make identity, biochemical and coarse positional evidence separately inspectable under controlled short-read perturbation.

Raw-sequence and position-aware models complete the picture. GeNet learns representations directly from raw DNA while exploiting the hierarchical label structure of metagenomic classification, and it shows that deep learning can reduce memory use without eliminating the need to compare against established mappers [36]. DeepVirFinder, Virtifier, and DETIRE demonstrate that CNN or hybrid architectures can extract useful motif- and composition-level signal from viral fragments, but their target is viral detection rather than multi-class short-read taxonomy [37, 38, 39]. KmerAperture adds an important boundary case by showing that ordinary k-mer set comparisons can lose synteny, whereas matching k-mers back to their original lists can retain positional information about core and accessory differences [40]. Collectively, these advances matured each representation family in isolation, but they still do not provide a unified diagnostic framework that decomposes ultra-short-read representations into identity, biochemical, and positional channels under controlled perturbation.

Against this background, we study short-read representation as a representation-audit problem rather than as a single-model accuracy competition. We ask how exact identity retention, biochemical perturbation stability, positional readability and compactness trade off across controlled short-read settings. Our compact main method, CK4P-MSP, combines reverse-complement canonical 4-mer identity with biochemical summaries and multi-scale property pooling to produce a low-dimensional, block-decomposable audit coordinate system. Full-position matrices serve as positional upper bounds that reveal what fine-grained positional resolution can add, whereas the canonical spaced-property (CSP) control is retained as a mechanistic boundary comparator for testing whether matching-oriented spaced-seed priors transfer to dense read-level representations. The study also incorporates same-dimension reduction controls, paired non-parametric significance tests, block-weight audits of mixed feature spaces, k-nearest-neighbor mutual-information robustness checks, dimension-matched high-k compressed k-mer baselines and local mutation fraction sweeps. The aim is therefore to clarify what each representation family preserves, what it obscures and where its useful operating boundary lies under a compact, interpretable feature budget.

Materials and Methods

Study design

We designed the study as a representation-diagnostic analysis of controlled short-read perturbation settings. The central question was what information remains inspectable in short-read sequence representations after controlled perturbation. We specifically asked whether biochemical and coarse positional summaries add auditable evidence beyond an exact identity backbone. CK4 and CK5 were used as reverse-complement canonical k-mer identity baselines, P denoted global biochemical-property summaries, MSP denoted multi-scale positional property pooling, CK4+P, CK4+MSP and CK4P-MSP tested compact mixed representations, full-position matrices were treated as positional upper bounds, and the canonical spaced-property (CSP) control served as a spaced-seed mechanism comparator.

This design makes two questions testable. First, whether a compact mixed representation remains close to its clean counterpart under nuisance perturbation while retaining representation-level readability. Second, whether any apparent benefit of the P or MSP channels can be separated from dimensionality, normalization or dilution in a mixed L2 space. We therefore evaluated compact stability, same-dimension PCA/SVD controls, dimension-matched high-k compressed k-mer baselines, block-weight sensitivity, paired non-parametric tests, k-nearest-neighbor mutual-information robustness checks, P/MSP contribution and counterfactual controls, P/MSP redundancy, feature-extraction runtime and error-aware ART strata as parts of one evidence chain. Shallow readout probes were used to test whether signal remains linearly accessible, not to define the main performance objective. Throughout, L2 in a mixed representation is interpreted as a standardized diagnostic drift after block construction and normalization, not as a natural biophysical distance between commensurate physical units.

The 69 and 75 bp settings were chosen because they fall within the commonly reported 50-75 bp post-QC mNGS read-length regime described in the interpretation literature (Yang, *Practice and Progress of mNGS Report Interpretation*, p. 99). These lengths were used as lower-read-length conditions in which identity evidence has less redundancy than at 150 bp, not as a claim that short reads are unusable. No patient-level reads, labels or sequence content from local restricted sequencing records were analyzed; those records informed only the choice of representative lower-read-length conditions. The remaining lengths were selected for sensitivity analysis: 100, 125 and 150 bp test whether the trends persist as reads become less extreme, and 300 bp is used only as an extended-read reference in selected boundary analyses.

Representation families

The representation families were defined to separate identity, global biochemical summaries and coarse positional information. CK4 and CK5 count reverse-complement canonical contiguous k-mers. P summarizes hydrogen-bond class, GC content, purine content, electron-ion interaction potential (EIIP)-like values, N fraction, entropy and length-related properties at the read level. MSP applies per-base property channels over relative-position bins. Its purpose is not to create a larger sequence vocabulary or to reconstruct full position. Instead, it adds a coarse positional property summary to an otherwise order-poor k-mer frequency vector. The default 2+3+4+6 binset was chosen as a coarse-to-fine relative-position summary whose finest scale remains above single-position resolution across the 69-150 bp regime. CK4+P concatenates canonical 4-mer identity with global property summaries, CK4+MSP tests positionalized property pooling without the global P block, and CK4P-MSP combines all three blocks. Full-position identity or property matrices retain per-position information and were used only to estimate an upper bound on positional readability. CSP combines spaced-seed counts with property summaries and was used to test whether matching-oriented spaced-seed priors transfer into dense representation diagnostics.

For mixed representations, each block was computed separately and internally L2-normalized before weighted assembly. For read s_i, let $\mathbf{K}_i$, $\mathbf{P}_i$ and $\mathbf{M}_i$ denote the reverse-complement canonical 4-mer identity block, the global biochemical-property block, and the multi-scale positional-property block, respectively. The internally normalized blocks were defined as

$$\hat{\mathbf{K}}_i = \frac{\mathbf{K}_i}{\|\mathbf{K}_i\|_2}, \quad \hat{\mathbf{P}}_i = \frac{\mathbf{P}_i}{\|\mathbf{P}_i\|_2}, \quad \hat{\mathbf{M}}_i = \frac{\mathbf{M}_i}{\|\mathbf{M}_i\|_2},$$

with zero-norm safeguards applied before normalization. The assembled compact representation was then

$$\mathbf{x}_i = \frac{1}{\sqrt{\alpha^2+\beta^2+\gamma^2}} \left[\alpha\hat{\mathbf{K}}_i \,;\, \beta\hat{\mathbf{P}}_i \,;\, \gamma\hat{\mathbf{M}}_i\right],$$

where α, β and γ are block weights and $[\cdot\,;\,\cdot]$ denotes concatenation. Unless otherwise stated, all three weights were set to 1. The block-weight audit varied these weights to test whether the observed pattern depended on a single identity/property scaling. Because the block norms are fixed before concatenation, the denominator is a fixed scalar for a given weight setting rather than a read-specific global norm. The reported global drift metric was

$$d(i,j) = \|\mathbf{x}_i - \mathbf{x}_j\|_2,$$

which should be interpreted as a standardized diagnostic drift in the assembled feature space, not as a natural physical distance between commensurate biological units. This definition also gives the block-normalized distance identity

$$\begin{aligned} d^2_{\mathrm{mix}}(i,j) = \frac{1}{\alpha^2+\beta^2+\gamma^2} \Big[&\alpha^2 \left\|\hat{\mathbf{K}}_i - \hat{\mathbf{K}}_j\right\|_2^2 \\ &+ \beta^2 \left\|\hat{\mathbf{P}}_i - \hat{\mathbf{P}}_j\right\|_2^2 \\ &+ \gamma^2 \left\|\hat{\mathbf{M}}_i - \hat{\mathbf{M}}_j\right\|_2^2 \Big]. \end{aligned}$$

For comparison with the identity-only drift $d_K(i,j) = \|\hat{\mathbf{K}}_i - \hat{\mathbf{K}}_j\|_2$, the mixed squared drift is lower than d_K^2 when

$$\beta^2\left(d_P^2 - d_K^2\right) + \gamma^2\left(d_M^2 - d_K^2\right) < 0,$$

where d_P and d_M are the corresponding blockwise drifts. Thus, lower property and MSP drift than identity drift is a sufficient, but not necessary, condition for a lower mixed drift. This explicit block construction allowed us to test whether the observed gains persisted when identity and property weights were changed. The design therefore does not assume that concatenation is intrinsically useful; it treats concatenation as a hypothesis that must survive same-dimension baselines, dimension-matched high-k compressed baselines, block-weight audits, blockwise drift decomposition and P/MSP counterfactual controls.

Data layers and perturbation design

The experiments were organized into eight data layers, with the scale of each layer reported in Table 2. The WGS perturbation layer contained 25,200 rows from 3,600 clean templates across six genera, 21 species and six read lengths. The position-property ablation layer expanded this to 36,000 rows over the same template set and length grid. ART Illumina-like simulation contributed 75,592 paired rows from 37,796 templates across 69-150 bp reads. CAMI_TOY_low probes used a 104,166-row initial labelled subset and a 216,000-row expanded subset with 30 labels at 69, 75 and 100 bp. The CAMI II marine lightweight probe used 4,000 anonymous source reads and 48,000 length/condition rows at 69, 75 and 100 bp. The local mutation layer used 250 triplets per analysis cell across 12 representations, four local modes and three read lengths. WGS-derived paired perturbations measured clean-versus-perturbed stability. ART Illumina-like simulation tested whether the same trends persisted under simulator-derived sequencing errors. A quality-stratified ART audit further split simulator outputs by read-quality strata, providing an error-aware check without claiming to model all clinical error processes. CAMI_TOY_low probes tested external metagenomic readability, whereas the CAMI II marine subset tested perturbation stability on anonymous external metagenomic reads without read-level taxonomic labels in this lightweight analysis. Controlled position and order tasks tested whether positional information was recoverable. Local mutation sensitivity compared structured local changes with matched nuisance perturbation. Boundary probes tested spaced-seed transfer, context visibility and ARG/SNP limits.

For CAMI_TOY_low, labels were normalized to a species-level view when available, and genus-level and target-versus-background probes were derived from the same normalized label field. Background or unclassified reads were retained only for probes that explicitly required a target/background split, whereas the labelled subsets were used for lightweight external readout at 30 labels across 69, 75 and 100 bp. The initial labelled subset supported the first external probe, and the expanded subset was used when the probe required broader label coverage. This lightweight CAMI use was intended to test whether representation-level readability survives an external metagenomic probe, not to approximate production-scale community profiling.

We also added a lightweight CAMI II marine subset as an external short-read stability probe. We streamed a prefix of CAMI II marine short-read sample 0 and parsed 4,000 anonymous 150 bp reads from the embedded `anonymous_reads.fq.gz` member. These reads were truncated to 69, 75 and 100 bp and expanded into clean, `N_3pct`, `substitution_1pct` and `substitution_1pct_N_3pct` conditions, yielding 48,000 length/condition rows. CAMI II provides benchmark truth resources, but the anonymous FASTQ headers did not provide read-level taxonomic labels in the streamed-read file used here, and the separate OTU-specific truth bundles were not reconstructed for this lightweight analysis. This CAMI II probe therefore reports paired perturbation-stability metrics on an external metagenomic sequence source rather than CAMI II taxonomic validation.

This layered simulation design was used because each component isolates a different failure mode. The controlled WGS grid makes perturbation type, read length and representation family directly comparable. ART adds a recognized Illumina-like simulator layer, but it is retained as a consistency check rather than treated as ground-truth clinical sequencing physics. The quality-stratified ART audit adds a more explicit error-aware view of the same simulator-derived layer. It supports statements about robustness across quality strata, but it does not by itself establish validation on clinical sequencing data.

Metrics and statistical analysis

Perturbation stability was quantified by paired cosine similarity, paired L2 drift and nearest-clean retrieval. Representation readability was quantified by macro-F1 and accuracy from shallow probes, mainly logistic regression and nearest-centroid classification. These readout metrics were used as accessibility probes for low-dimensional features, not as claims of production classifier performance. Compactness was measured by feature dimension. Local mutation analysis used selective-sensitivity ratios and delta-readout to distinguish nuisance stability from sensitivity to structured local change. Bootstrap intervals summarized cell-level variation. Paired Wilcoxon tests with Benjamini-Hochberg correction were used for matched stability comparisons. Same-dimension PCA/SVD controls addressed dimensionality effects, and block-weight audits tested whether mixed-feature conclusions were stable under reasonable reweighting of identity and property blocks.

The mixed-feature question was also tested through empirical information and counterfactual audits. For a perturbation label Y and representation-derived distance or readout features X, we estimated discretized pooled and conditional MI proxies using equal-frequency binning. The main pooled distance audits used eight bins, joint P/MSP summaries used five- or six-bin discretizations depending on the audit, and empirical permutation baselines used 200 shuffles per cell. These values are reported as estimator-dependent descriptive proxies rather than absolute physical information quantities. To test estimator sensitivity, we also ran a nearest-neighbor MI robustness audit using Ross/Kraskov-Stogbauer-Grassberger (KSG)-style estimators for continuous distance summaries and a discrete local-versus-noise perturbation label. This audit used $k = 5$, 50 label permutations, 50 stratified subsampling repetitions and low-dimensional summaries including d_K, d_P, d_M, $d_K + d_P$, $d_K + d_M$, and $d_K + d_P + d_M$ across the 69, 100 and 150 bp local-mutation cells. To separate the global and positional property layers, we additionally evaluated CK4, CK4+P, CK4+MSP and CK4P-MSP under the same block-normalized assembly rule. This direct contribution audit measured paired stability and local delta-readout, and was paired with a redundancy audit based on rank, canonical-correlation analysis and row-wise P/MSP alignment. To test whether the P/MSP contribution was more than additional dimensions, we also constructed CK4 plus permuted P/MSP blocks and CK4 plus Gaussian P/MSP blocks. These controls preserve a larger feature space but remove the sequence-linked biochemical mapping. The analysis was used as empirical support for perturbation-associated signal in the added property layers, not as a dataset-independent information-theoretic guarantee or as evidence that P and MSP measure unrelated physical quantities.

We added a dimension-matched high-k compression audit to avoid comparing CK4P-MSP only with short contiguous k-mer baselines. Canonical $k = 15$ signals were compressed to approximately the 222-feature CK4P-MSP budget using three controls: MinHash-style signatures, hashing-trick counts and sparse random projection. These baselines were evaluated on the same 69, 75, 100 and 150 bp perturbation grid using paired cosine similarity, L2 drift, nearest-clean retrieval and shallow readout. This audit tests whether the stability of CK4P-MSP persists against high-specificity k-mer information under a comparable compactness constraint. Finally, a feature-extraction runtime audit compared CK4P-MSP with CK4, CK4+P and dimension-matched high-k compressed controls. Runtime was treated as an engineering boundary, not as a primary performance claim.

Reproducibility

All generated results used local scripts and result tables stored under the project workspace. Main figure assets are in `info/paper/figures`, table assets are in `info/paper/tables`, the baseline supplementary figures are generated by `info/scripts/generate_paper_supplementary_figures.py`, the kNN MI robustness audit by `info/scripts/run_knn_mi_robustness_audit.py`, the high-k compressed baseline audit by `info/scripts/run_high_k_compressed_baselines.py`, the P/MSP contribution audit by `info/scripts/run_p_msp_contribution_audit.py`, the P/MSP redundancy and runtime audit by `info/scripts/run_property_redundancy_and_runtime_audit.py`, the P-channel counterfactual audit by `info/scripts/run_p_channel_counterfactual_audit.py`, the CAMI II marine lightweight probe by `info/scripts/run_cami2_marine_lightweight_probe.py`, and its supplementary figure by `info/scripts/generate_supp_fig_s10_cami2_marine_probe.py`. The supported scope is compact and controlled: biochemical and position-aware summaries are evaluated as block-decomposable representation-level audit coordinates in short-read settings.

Results

Empirical mixed-feature information audit under local perturbation

We first tested whether the added biochemical channel showed perturbation-associated signal rather than simply changing the scale of a mixed feature space. In the local-mutation audit, the equal-frequency MI proxy gave a higher pooled value for joint CK4+P/MSP distances than for CK4 distance alone, and the conditional proxy remained positive after conditioning on CK4 distance. Because discretized MI estimates are estimator-dependent, we treated this result as a descriptive screen and repeated the analysis with a k-nearest-neighbor MI robustness audit on low-dimensional distance summaries. Across the 12 local-mutation cells, the kNN audit estimated higher MI for the joint $d_K + d_P + d_M$ summaries than for d_K alone, with a mean incremental estimate of approximately 0.138 bits beyond d_K and permutation support across the tested cells. This result is reported as estimator-dependent empirical evidence: on the tested perturbation grid and estimators, P/MSP distances showed perturbation-associated signal beyond a pure identity-only distance.

This audit is important for the interpretation of CK4P-MSP. A lower standardized drift could otherwise be attributed to dimensional dilution by low-variance auxiliary features. The MI robustness checks support a more specific conclusion: the property and MSP channels contributed estimator-dependent perturbation-associated signal under the tested local-change regime, while the block-weight audit separately checked that the conclusion was not tied to one arbitrary block scaling. The claim is therefore metric-based and empirical, not a statement that heterogeneous L2 coordinates have identical physical units.

See Supplementary Figure S2.

See Supplementary Figure S7.

See Supplementary Figure S5.

Global and positional property layers have related but distinct roles

We then separated the two property layers directly. In a matched contribution audit over 69, 75, 100 and 150 bp reads, CK4 alone had a mean paired L2 drift of 0.254. Adding either global P or MSP reduced this drift to 0.182 and 0.181, respectively, and CK4P-MSP reduced it further to 0.150. Thus, both property layers contributed to nuisance-perturbation stability under the block-normalized metric.

The local delta-readout pattern was more specific. CK4 and CK4+P achieved mean macro-F1 values of 0.888 and 0.894 for distinguishing structured local change from matched nuisance perturbation, whereas CK4+MSP reached 0.976 and CK4P-MSP reached 0.977. This indicates that the positionalized property layer, rather than the global P block alone, carried most of the coarse positional readout gain. This supports the intended design: P provides a global biochemical stability summary, whereas MSP uses the same property family to add low-dimensional positional property summaries to a compact identity-frequency backbone.

The redundancy audit also cautions against overinterpreting P and MSP as separable sources. Across the tested lengths, the first canonical correlation between P and MSP was near one and their first principal components were strongly aligned, as expected because both blocks are derived from the same biochemical property family. However, MSP had higher numerical rank than P, and row-wise P/MSP alignment after compression was incomplete. We therefore interpret the blocks as related layers at different scales, not as interchangeable evidence sources or as measurements of unrelated physical quantities.

See Supplementary Figure S8.

See Supplementary Figure S9.

MSP short-bin reliability remains usable at 69-75 bp

We next tested whether MSP becomes noise-dominated when ultra-short reads are divided into finer relative-position bins. The short-bin audit did not show an immediate loss of reliability at 69 or 75 bp. In the stability screen, moving from a 2-bin MSP to the full 2+3+4+6-bin MSP changed paired cosine and L2 drift only slightly at the default $\gamma = 1$ setting: at 69 bp, paired cosine remained approximately 0.986 and L2 drift remained near 0.147; at 75 bp, paired cosine remained approximately 0.987 and L2 drift remained near 0.141. Retrieval also remained effectively saturated.

The local delta-readout audit did not show an immediate short-bin failure. Finer multi-scale pooling increased mutation readout at fixed gamma. At 69 bp and $\gamma = 1$, mean macro-F1 increased from approximately 0.865 with 2 bins to approximately 0.899 with 2+3 bins, 0.946 with 2+3+4 bins and 0.953 with 2+3+4+6 bins. The same monotonic pattern persisted at 100 and 150 bp. Bootstrap widths for pooled property features remained small, whereas worst-case binomial sampling error increased as expected when bins narrowed. The combined interpretation is therefore restrained but favorable: static MSP did not become noise-dominated in the tested ultra-short-read regime, yet it should still be interpreted as a coarse positional summary rather than as an optimized variance-aware weighting model.

See Supplementary Figure S6.

Error-aware ART perturbation audit

We next asked whether the external simulator evidence depended on treating ART as one pooled error source. In the quality-stratified ART audit, representative compact controls remained readable across low-, mid- and high-quality bins. The property-aware compact representation preserved high paired cosine and bounded L2 drift across quality strata, and the hybrid identity/spaced representation retained retrieval and readout consistency.

These results strengthen, but do not overstate, the ART evidence. They show that the main pattern is not restricted to an undifferentiated simulator summary. They do not claim that ART quality strata capture the full physical error distribution of a clinical sequencing run.

See Supplementary Figure S3.

Compact biochemical summaries reduce perturbation drift

The first main experimental block tested whether compact property-aware representations remained closer to their clean counterparts than identity-only baselines. Across the WGS-derived perturbation grid, CK4P-MSP and related property-aware compact representations retained high paired cosine and lower L2 drift while preserving nearest-clean retrieval. In the compact summary, CK4P-MSP achieved paired cosine around 0.991 with L2 drift around 0.116 at 222 features, whereas identity-only compact baselines showed higher drift under the same perturbation families.

The same conclusion survived several fairness checks. Same-dimension PCA/SVD controls showed that the mixed compact result was not simply a consequence of using more features. Dimension-matched high-k compression controls further addressed whether a conventional high-specificity k-mer representation could show the same behavior under the CK4P-MSP feature budget. In that audit, CK4P-MSP retained higher paired cosine and lower L2 drift than $k = 15$ random projection, MinHash and hashed baselines at approximately 222 features (paired cosine 0.996 and L2 drift 0.078 for CK4P-MSP, compared with paired cosine 0.848, 0.841 and 0.794 and L2 drift 0.484, 0.506 and 0.572 for the three compressed high-k controls). Shallow readout differences were modest (logistic macro-F1 0.401 for CK4P-MSP versus 0.396 for the hashed $k = 15$ control), so this result supports a compact perturbation-stability and auditability advantage rather than a predictive-performance claim. Paired Wilcoxon tests over matched analysis cells supported the stability differences for key comparisons after multiple-testing correction. Block-weight audits showed that stability and readout conclusions did not depend on one fixed identity/property weighting. P/MSP counterfactual controls further separated dimensionality from sequence-linked biochemical information: CK4 plus permuted P/MSP approached the stability of CK4P-MSP but did not reproduce its local delta-readout (macro-F1 0.875 versus 0.977), while CK4 plus Gaussian P/MSP collapsed both stability and readout. The P/MSP contribution audit further showed that CK4+P and CK4+MSP both reduced drift relative to CK4, while MSP accounted for most of the positional delta-readout gain. Together, these controls show that the stability trend is not explained by feature count, a single weighting choice or random auxiliary columns.

See Supplementary Figure S1.

See Supplementary Figure S7.

CK4P-MSP balances compactness, stability and representation readability

We then evaluated CK4P-MSP as a nested ablation rather than as a predictive model. CK4 tested compact identity alone. CK4+P tested whether global biochemical summaries were sufficient. CK4+MSP tested whether positionalized property pooling carried information beyond identity without the global P block. CK4P-MSP tested the combined trade-off between stability, dimensionality and shallow readout when global and positional property layers were added together. This hierarchy is central to the method: the representation is intended to preserve exact identity evidence while adding interpretable biochemical and coarse positional audit channels.

The resulting trade-off supported the intended role. CK4P-MSP remained compact, reduced perturbation drift relative to CK4 and CK5 identity baselines, and retained accessible signal in shallow readout. Its role is to provide a block-decomposable audit coordinate system between identity-only compact features and high-dimensional full-position matrices, rather than to maximize predictive accuracy against learned embeddings.

External ART and CAMI probes separate stability from task-limited readability

ART and CAMI were used to test whether the representation trend survived outside the handcrafted perturbation grid, while keeping stability and readout as separate questions. ART provided simulator-derived sequencing-error evidence: in the paired-cosine summary, property-aware and spaced-property representations retained higher clean-versus-perturbed similarity than the identity-only k-mer comparators (Figure 4A). CAMI_TOY_low provided a lightweight external metagenomic readout probe with two different granularities. A coarse target-versus-background task remained readable across compact representations (Figure 4B), whereas the 30-label fine probe was substantially lower in absolute macro-F1 (Figure 4C). Thus, the external CAMI result supports representation-level readability at a coarse task level, but it also shows that fine-grained label readout remains task-limited and should not be interpreted as production-grade taxonomic classification.

The quality-stratified ART audit complements this result by showing that simulator consistency persisted across read-quality bins. To test whether the compact-stability trend was restricted to CAMI_TOY_low, we added the CAMI II marine anonymous-read subset probe (Supplementary Figure S10). Across all nine combinations of length (69, 75 and 100 bp) and perturbation (`N_3pct`, `substitution_1pct` and `substitution_1pct_N_3pct`), CK4P-MSP had the lowest mean L2 drift among the tested compact representations and retained nearest-clean retrieval. Mean paired cosine for CK4P-MSP was approximately 0.995 under N masking, 0.998 under substitution and 0.992-0.993 under the combined perturbation, with mean L2 drift ranges of 0.095-0.102, 0.047-0.048 and 0.114-0.121, respectively. Marine metagenomes may differ from clinical or pathogen-rich contexts in sequence composition, so this result should be interpreted as an external sequence-source stability probe. It supports the external stability pattern in a more complex CAMI II metagenomic read source, while remaining a stability probe without read-level taxonomic labels in this lightweight analysis rather than a taxonomic validation.

Together, these analyses support external consistency under controlled simulator and benchmark resources, while preserving the manuscript boundary: ART supports simulator-derived stability consistency, CAMI_TOY_low supports external readability at a task-limited level, and CAMI II marine supports anonymous-read stability rather than taxonomic validation.

See Supplementary Figure S10.

Full-position matrices define a positional upper bound rather than a deployment default

Full-position encodings were used to estimate what is gained when fine-grained layout information is retained. In controlled position and order tasks, full-position identity or property matrices improved positional readability compared with compact summaries. This was expected, because these matrices preserve per-position structure that compact summaries intentionally compress.

The result is best interpreted as an upper-bound diagnostic. Full-position matrices clarify the value of positional information, but their high dimensionality and task specificity make them unsuitable as the default representation for compact short-read auditing. CK4P-MSP therefore occupies a different role: it does not match the full positional upper bound, but it retains a coarse positional property summary at a much smaller feature cost.

Local mutation analysis separates robustness from selective sensitivity

A stable audit representation should remain responsive to structured changes. We therefore tested whether local biochemical change remained distinguishable from matched nuisance perturbation. Property-aware and full-position channels showed larger response to structured local mutation than to matched noise in the relevant comparisons, while CK4P-MSP preserved delta-readout at compact dimensionality. The P/MSP counterfactual audit sharpened this result by showing that the real biochemical mapping, not only extra columns, was needed to obtain the strongest local-change readout.

This result supports a division of labor. Paired cosine and L2 drift quantify nuisance stability, whereas selective-sensitivity ratios and delta-readout quantify whether structured local changes remain readable. CK4P-MSP is therefore not merely a smoothed representation. It can remain stable under nuisance perturbation while preserving decomposable information for local-change readout.

See Supplementary Figure S4.

Boundary analyses constrain the interpretation

The boundary probes were included to prevent overextension of the representation claim. Spaced-seed transfer was useful as a mechanism comparator, but matching-oriented seed priors did not automatically become a universal dense-feature advantage. Context visibility depended on whether the relevant relation was present within the read. ARG/SNP interpretation required identity, annotation and allele-level evidence beyond what compact property summaries can provide.

These negative and boundary results are part of the manuscript's logic. They keep the claim focused on representation diagnostics and representation-level perturbation auditing, rather than allowing the compact feature space to be misread as a full biological interpretation system.

Table 1. Representation families and diagnostic roles.

family	information channel	main role	main-text interpretation
CK4 / CK5	exact local identity	identity backbone and comparator	necessary for taxonomic, allele and SNP-level evidence
P	biochemical summaries	perturbation-stable side channel	adds robustness information but does not replace identity
CK4+P	identity plus global biochemical signal	global property baseline	tests whether biochemical summaries add to CK4
CK4+MSP	identity plus positionalized biochemical signal	positional property ablation	tests whether MSP returns coarse layout information without global P
CK4P-MSP	identity plus multi-scale biochemical layout	compact main representation	balances stability, readability and feature dimension
full-position matrix	per-position identity or property signal	upper-bound diagnostic	tests fine positional information at higher dimensional cost
CSP	spaced-seed prior plus biochemical summary	spaced-seed transfer comparator	delimits where matching-oriented spaced seeds transfer to dense features

Table 2. Data layers, diagnostic questions, metrics and claim boundaries.

data layer	scale	diagnostic question	primary metrics	claim boundary
WGS perturbation pairs	25,200 rows; 3,600 clean templates; 6 genera/21 species; lengths 69,75,100,125,150,300 bp	Which representations keep clean and perturbed reads close?	paired cosine, L2 drift, retrieval	controlled representation stability, not clinical validation
Position-property ablation	36,000 rows; 3,600 templates; 6 genera/21 species; lengths 69,75,100,125,150,300 bp	What do P and MSP add to the CK4 identity backbone?	paired stability, block drift, delta-readout, MI proxies	decomposable representation audit, not a natural physical metric
ART Illumina-like simulation	75,592 paired rows; 37,796 templates; 6 genera/21 species; lengths 69,75,100,125,150 bp	Do stability trends persist under simulator-derived read errors?	paired cosine, L2 drift, retrieval	platform-like consistency check
CAMI_TOY_low probes	104,166-row initial subset and 216,000-row expanded subset; 30 labels; lengths 69,75,100 bp	Is information readable in an external metagenomic readout?	macro-F1, accuracy	lightweight readout, not clinical classification
CAMI II marine anonymous-read probe	48,000 rows from 4,000 anonymous source reads; lengths 69,75,100 bp	Do stability trends persist in a more complex external metagenomic read source?	paired cosine, L2 drift, retrieval	paired stability without reconstructed read-level taxonomic labels
controlled position tasks	controlled synthetic tasks over the short-read length grid	What does fine positional resolution add?	macro-F1, feature dimension	upper-bound diagnostic
local mutation sensitivity	250 triplets per analysis cell; 12 representations; 4 local modes; lengths 69,100,150 bp	Can local biochemical change be detected beyond equal-count noise?	sensitivity ratio, delta-readout macro-F1	selective sensitivity, not functional effect prediction
boundary probes	spaced-seed, context-visibility and ARG/SNP boundary screens	Where do spaced-seed, context and ARG/SNP claims stop?	seed stability, visibility threshold, macro-F1	mechanism boundary and failure modes

Table 3. Compact main-method metrics.

representation	paired cosine	L2 drift	readout macro-F1	features
CK4	0.935	0.316	0.378	136
CK4+P	0.990	0.122	0.380	147
CK4P-MSP	0.991	0.116	0.401	222
CK5	0.882	0.433	0.373	512

Table 4. Local mutation sensitivity metrics.

representation	noise L2	local L2	local − noise L2	sensitivity ratio	delta-readout macro-F1	features
CK4	0.355	0.266	-0.089	0.750	0.891	136
CK4+P	0.115	0.086	-0.029	0.752	0.900	147
CK4P-MSP	0.110	0.083	-0.027	0.754	0.977	222
CSP	0.116	0.109	-0.006	0.947	0.696	147
one-hot	0.239	0.239	0.000	1.000	0.956	500
P-channels	0.086	0.107	0.020	1.255	0.983	500
one-hot+P	0.188	0.207	0.019	1.101	0.973	1000
position k-mer P	0.167	0.185	0.017	1.125	0.972	1536

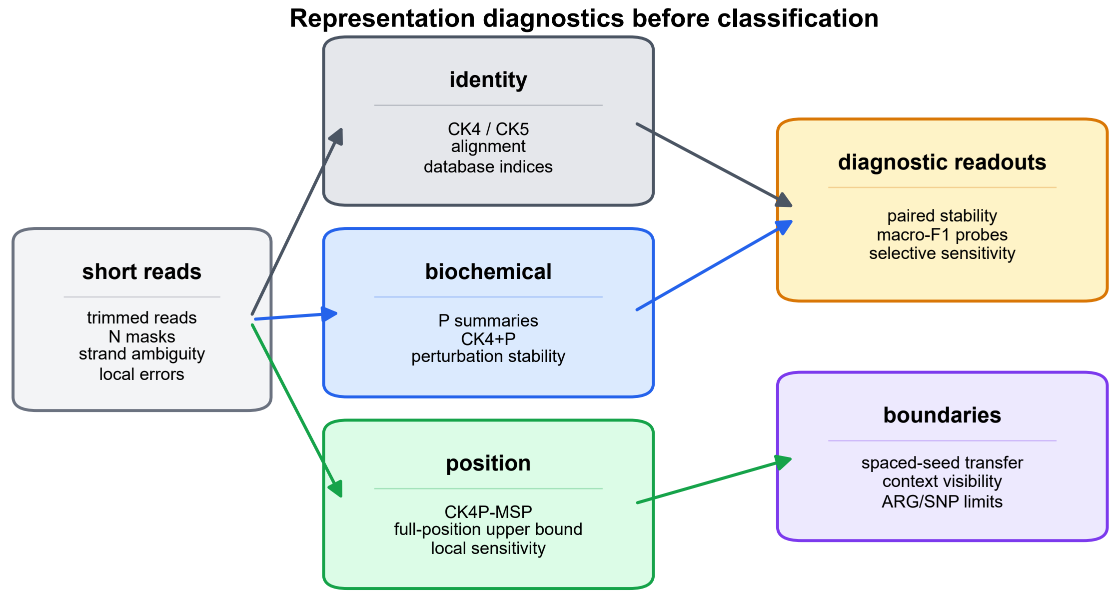

Figure 1 Representation-diagnostic framework and read-length regime.

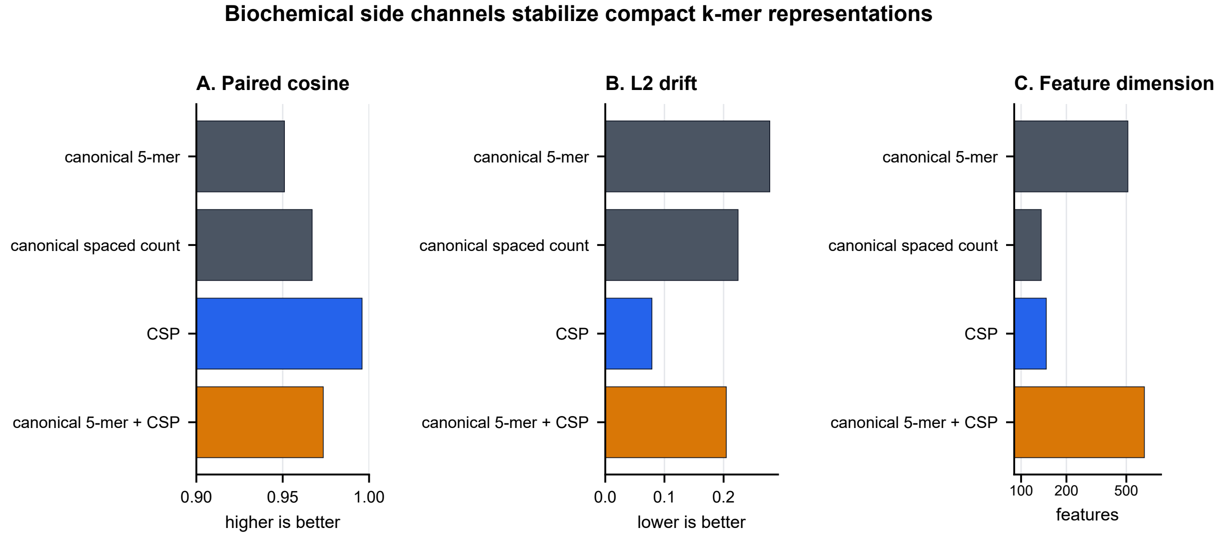

Figure 2 Compact stability under controlled perturbation.

Table 5. Boundary and mechanism summary.

boundary	main observation	interpretation
spaced-seed transfer	The best tested 4-position dense stability pattern was contiguous 0-1-2-3 across the 69/75 bp sanity grid.	spaced seeds remain useful matching priors, but transfer is regime-specific
context visibility	motif-pair visibility thresholds depended on placement, emerging at 130, 140, 148 or 155 bp.	a representation cannot encode context absent from the read
ARG/SNP readout	stable compact features did not by themselves establish allele, resistance-SNP or functional equivalence.	identity/database evidence remains necessary for final biological calls

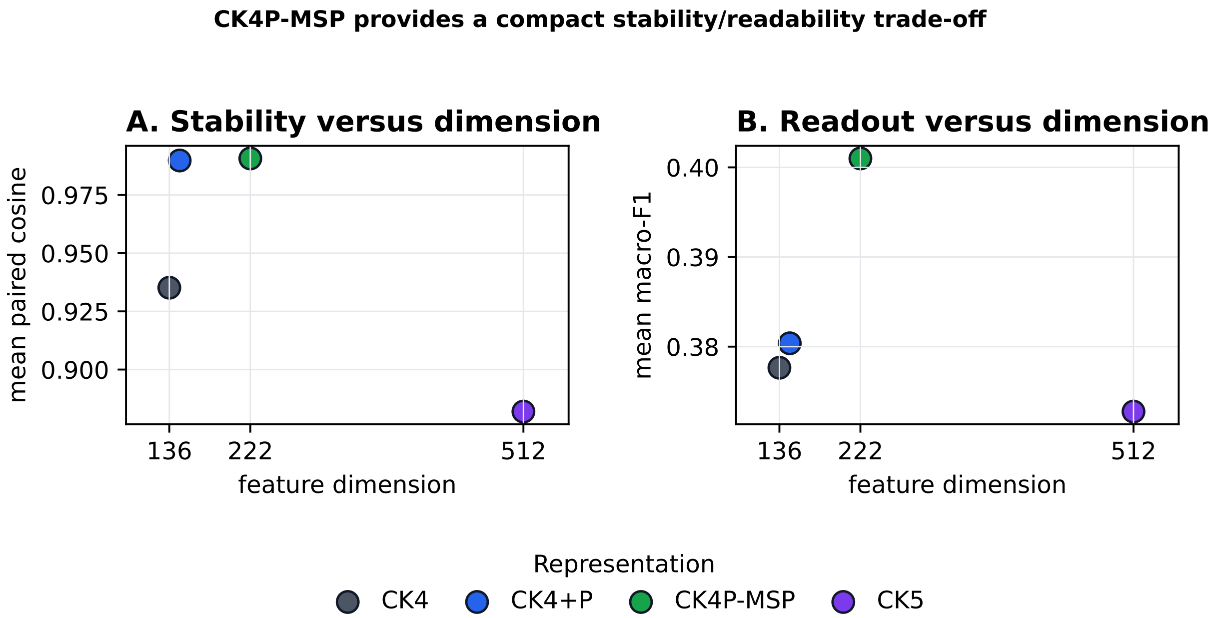

Figure 3 CK4P-MSP stability-readout-dimension trade-off.

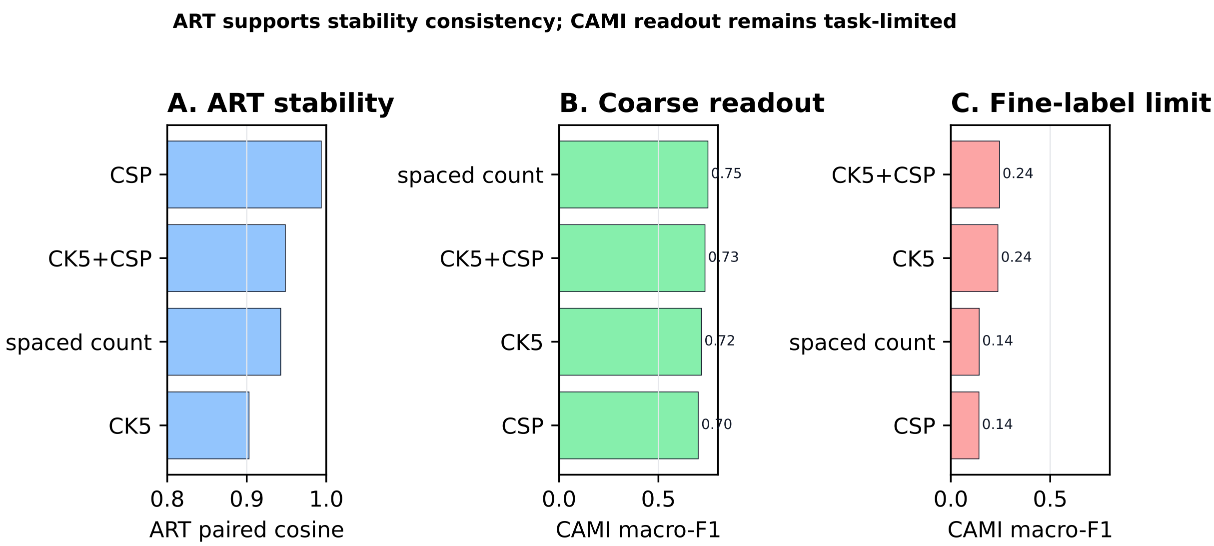

Figure 4 ART stability and CAMI coarse/fine readout probes.

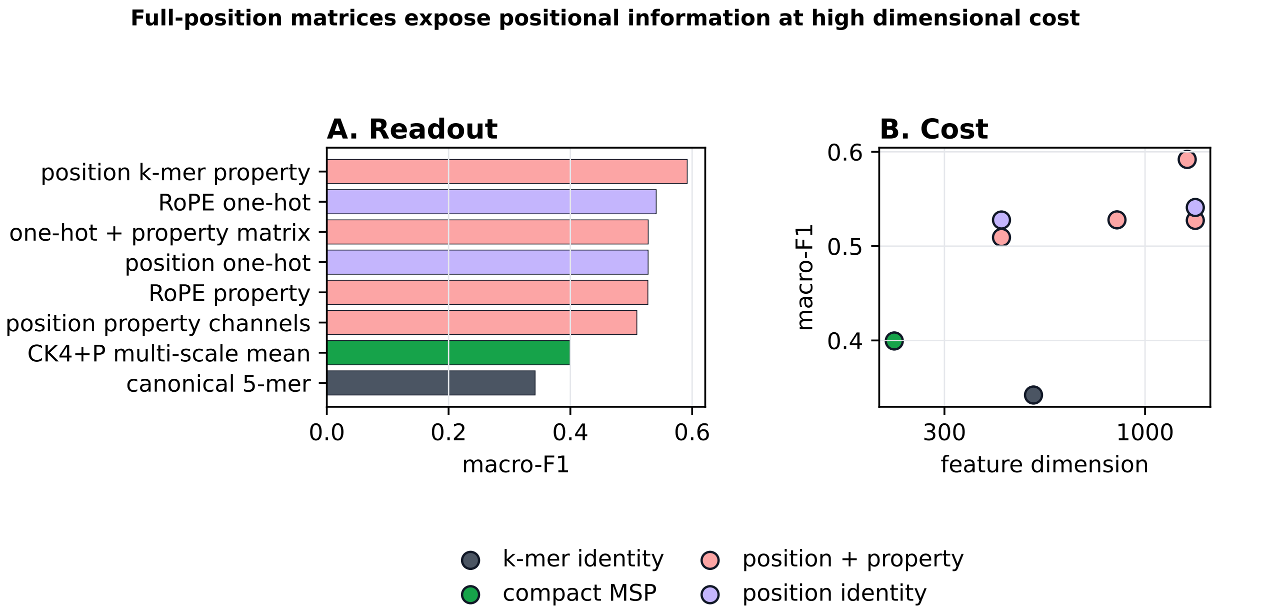

Figure 5 Full-position diagnostic upper bound for positional information.

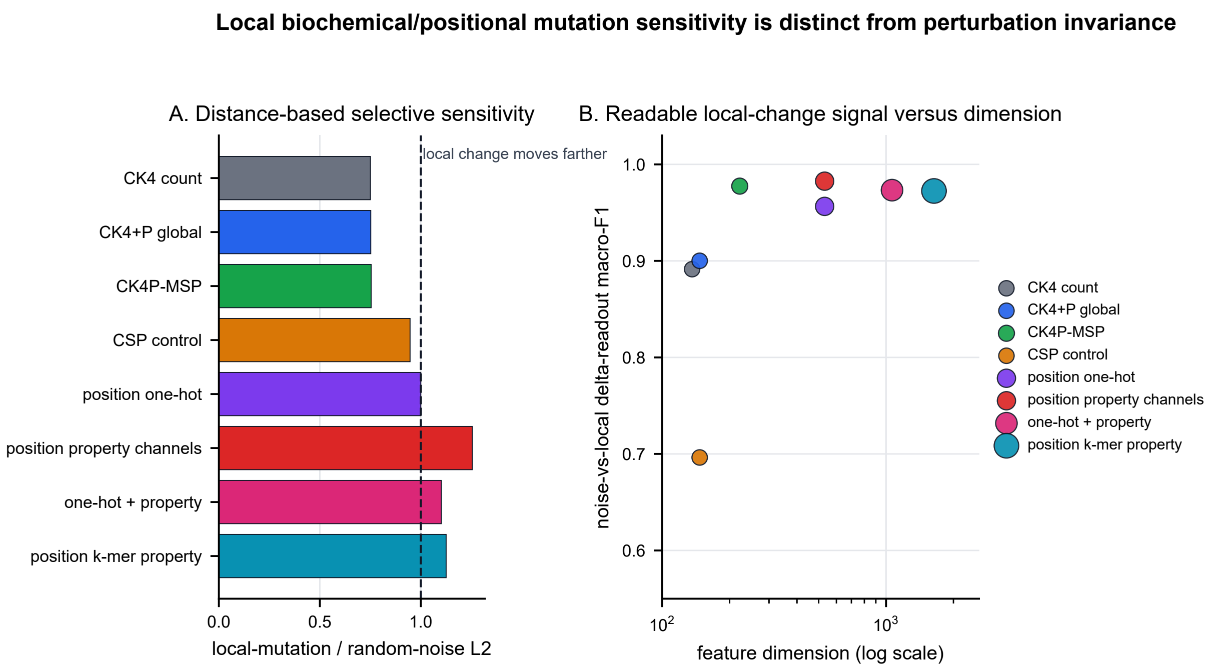

Figure 6 Local mutation sensitivity and delta-readout.

Discussion

This study treats short-read sequence representation as an inspectable object rather than only as a black-box prelude to classification. Across controlled short-read settings, exact identity evidence remained essential, but biochemical and coarse positional summaries changed what could be measured under perturbation. CK4P-MSP provided the clearest compact trade-off in this role: it preserved a reverse-complement canonical 4-mer identity backbone while adding biochemical and multi-scale property pooling channels that reduced perturbation drift and retained representation-level readability at compact dimensionality.

The results support a division of labor between representation families. Exact k-mer features provide the identity backbone needed for database-linked interpretation. Global biochemical summaries provide a stability channel that can remain informative when individual exact matches are weakened by trimming, ambiguity or local perturbation. Multi-scale positional property pooling uses the same biochemical property family to add coarse layout summaries to an otherwise order-poor k-mer frequency vector. It is not a larger k-mer vocabulary, a learned token mapping or a reconstruction of full positional structure. Full-position matrices, by contrast, define what can be gained when positional information is preserved almost explicitly. They are useful diagnostic upper bounds but are not the preferred compact representation.

The additional analyses sharpen this interpretation. Same-dimension PCA/SVD controls show that the compact mixed representation is not simply benefiting from a larger feature budget. Dimension-matched high-k compressed baselines show that the stability result is not limited to comparisons with short contiguous k-mer counts: CK4P-MSP remained more stable than $k = 15$ MinHash, hashed and random-projection controls at a comparable feature budget, although shallow readout differences were modest. Block-weight audits show that the conclusion is not tied to one arbitrary identity/property scaling. Paired Wilcoxon tests verify that the main stability contrasts are reproducible across matched analysis cells. The empirical MI and k-nearest-neighbor MI audits indicate that P/MSP distance summaries showed perturbation-associated signal beyond CK4 distance in the tested local-mutation regime.

The direct P/MSP contribution audit showed that both CK4+P and CK4+MSP reduced drift relative to CK4, whereas MSP carried most of the local delta-readout gain. The redundancy audit further showed that P and MSP share a strong global property component but are not interchangeable: MSP has higher numerical rank and incomplete row-wise alignment with P. The P/MSP counterfactual audit separates sequence-linked biochemical information from feature-space expansion. Permuted P/MSP blocks approach some stability behavior but do not reproduce the strongest delta-readout, whereas Gaussian auxiliary blocks fail both stability and readout. These analyses do not provide a general mathematical guarantee, but they make the interpretation less dependent on mixed-space dilution, feature count, redundancy or a single weighting choice.

The perturbation analyses also clarify the scope of the evidence. Uniform substitutions and local mutation sweeps provide controlled tests of representation behavior. ART adds a simulator-derived Illumina-like layer, and the quality-stratified ART audit shows that the main pattern persists across read-quality strata. CAMI_TOY_low provides a labelled external metagenomic readout probe, whereas the CAMI II marine subset provides a more complex anonymous-read external stability probe without read-level taxonomic labels in this lightweight analysis. Together, these data form a consistent set of evidence for representation-level stability and readability under the tested conditions. The manuscript reports the scale of each data layer so that controlled grids, simulator checks, labelled lightweight readout probes and anonymous-read stability probes are not conflated.

A useful outcome of the boundary analyses is that they prevent overgeneralization. CSP and spaced-seed controls help interpret which matching-oriented priors transfer into dense read-level representations, but they do not become general substitutes for compact identity-property features. Context and ARG/SNP probes show that compact stability is not the same as allele-level or functional interpretation. Those tasks require database identity, annotation, sufficient context and downstream biological evidence.

Limitations

The first limitation is task scope. The study evaluates representation diagnostics, whereas Kraken 2, Centrifuge, CLARK, Kaiju and related systems remain the appropriate reference frame for database-linked taxonomic assignment. The present work asks what information remains visible before or alongside such assignment systems. Direct integration with production classifiers, including downstream false-hit reduction or classifier-output auditing, remains future work and is not evaluated here.

The second limitation is the perturbation model and external benchmark scale. Controlled substitutions, ambiguous-base perturbations and local mutation sweeps are useful because they isolate representation behavior, but they do not reproduce every physical process in sequencing data. ART and the quality-stratified ART audit partly address this issue by adding simulator-derived and quality-aware evidence. CAMI_TOY_low and CAMI II marine add external metagenomic resources, but with different roles: the former supports lightweight labelled readout, and the latter supports only paired stability because read-level taxonomic labels were not reconstructed for the anonymous reads in this lightweight probe. The resulting conclusions therefore remain bounded to controlled short-read, simulator-supported and external stability/readout settings, not to production-scale community profiling or unobserved clinical error distributions.

The third limitation concerns data provenance. The 69 and 75 bp settings were selected as lower-read-length conditions that fall within a commonly reported post-QC mNGS range, and local restricted sequencing records were not used as study inputs or publicly shared. The experiments therefore use length conditions, not patient-level reads or patient-derived labels. The final submission will include institutionally approved wording for this data-governance boundary.

The fourth limitation is that the MI and conditional-MI analyses are empirical. They support perturbation-associated signal in the tested feature channels, especially after the k-nearest-neighbor robustness audit, but they are not general evidence that a property-augmented representation always has higher mutual information than an identity-only representation. Likewise, the mixed L2 distance should be read as a standardized diagnostic drift after block construction and normalization, not as a natural metric that equates biochemical properties and k-mer counts as identical physical units. The correct claim is therefore conditional: under the tested local-perturbation grid and estimators, property-aware distance summaries showed additional mutation-relevant signal relative to identity-only and permutation controls.

The fifth limitation concerns MSP weighting and scale. The short-bin audit did not show an immediate loss of reliability at 69-75 bp, and finer multi-scale pooling increased delta-readout rather than immediately degrading it. Even so, the current MSP channel remains a static coarse summary. It is not an optimized variance-aware estimator, and the manuscript does not claim that its present binset or gamma weighting is universally optimal across read lengths, perturbation types or sequencing platforms.

The sixth limitation is computational scope. CK4P-MSP remained lightweight in absolute terms in the local implementation, but it was slower to extract than CK4 and simple high-k hashed or MinHash summaries. Its value in this manuscript is interpretability and perturbation diagnostics under compact dimensionality rather than engineering speed-up. Large-scale deployment would require optimized implementation and pipeline-level cost-benefit testing.

The seventh limitation is model scope. Large DNA language models and learned metagenomic classifiers provide powerful sequence representations. They may outperform CK4P-MSP in predictive probes, especially when sufficient training data, pretraining and compute are available. This does not contradict the present claim because the study's goal is different. CK4P-MSP is intended as a low-dimensional, training-free and block-decomposable coordinate system for auditing identity, biochemical and coarse positional information under controlled short-read perturbation. Direct comparison with learned embeddings would require a separate benchmark design that reports both predictive accuracy and auditability.

Future work

Future work should test whether CK4P-MSP or related compact summaries improve failure-mode detection when placed alongside database-centered pipelines. Pipeline-facing false-hit reduction or classifier-output auditing would require downstream classifier outputs and explicit operating-point analysis. The most useful next step is to add representation-level diagnostics that report when identity evidence is fragile, when biochemical stability remains high and when local-change sensitivity is preserved.

A second direction is to expand the perturbation model. Platform-specific error-transition matrices, explicit Q-score trajectories and larger simulator grids would make the error-aware analysis more realistic. The current controlled simulations are useful because they isolate representation behavior by read length, perturbation type and feature family; future work should test whether the same conclusions hold under richer platform-specific error processes. When ethically and legally feasible, restricted clinical read collections could then be used for external validation under controlled access, with public release limited to aggregate metadata and reproducible scripts.

A third direction is to make the positional channel more adaptive without changing the paper's current claim. Length-aware or variance-aware MSP weighting could be explored in future work, but such a step should be framed as a new model family rather than as a cosmetic parameter retune. The present results justify static MSP as a usable compact summary; they do not settle the separate optimization problem of how positional bins should be weighted under different noise and length conditions.

A fourth direction is to connect compact diagnostics with learned models. Full-position matrices and DNA foundation-model embeddings are natural candidates for follow-up studies because they may recover richer context and stronger predictive signal than compact summaries. The present work suggests where such models should be audited: not only by final classification accuracy, but also by identity retention, perturbation stability, positional readability and local-change sensitivity.

Conclusion

This study shows that compact short-read representations can be evaluated as interpretable audit objects. In controlled short-read settings, CK4P-MSP retained an exact identity backbone while adding global biochemical and positionalized biochemical channels that reduced perturbation drift and preserved local-change readability. The evidence supports a bounded implication: compact biochemical and coarse position-aware summaries can provide block-decomposable coordinates for representation-level perturbation auditing alongside exact matching and curated database methods.

Data Availability

All processed data tables, figure source summaries and analysis outputs generated for this study are maintained in a private GitHub repository for journal review (`https://github.com/FairYJmrx/ultrashort-dna-representation-diagnostics`, release branch). Reviewer access will be provided through the submission system or by adding a journal-supplied account as a read-only collaborator. A public archival DOI will be minted from the frozen release when the authors make the repository public.

Publicly reused resources include simulator- and benchmark-derived materials used for ART, CAMI_TOY_low and CAMI II marine analyses, which will be cited through the corresponding primary publications and public resource records in the final reference list. The CAMI II marine lightweight probe used the public short-read sample 0 archive (`https://frl.publisso.de/data/frl:6425521/marine/short_read/marmgCAMI2_sample_0_reads.tar.gz`) and setup archive (`https://frl.publisso.de/data/frl:6425521/marine/short_read/marmgCAMI2_setup.tar.gz`). Only a streamed prefix was parsed for the lightweight stability probe, and read-level taxonomic labels were not reconstructed from the separate truth resources for this analysis. The generated subset tables, stability summaries and source figure table are included in the review-access repository and will be included in the public release. Local restricted sequencing records were used only as read-length provenance for the 69 and 75 bp conditions. The underlying clinical sequencing reads were not used as experimental input, are not part of the study data package, and cannot be publicly shared.

Code Availability

The analysis scripts used to generate tables, figures and reviewer-response audits are maintained in the private review-access repository (`https://github.com/FairYJmrx/ultrashort-dna-representation-diagnostics`, release branch) under an MIT licence. The repository README and release manifest map each manuscript figure, table and supplementary audit to its source script, input tables and generated outputs. Reviewer access will be provided during submission, and a public archived release identifier will be added after public release.

Ethics and Data Governance

No patient-level clinical sequencing reads, patient identifiers or patient-derived labels were analyzed in this study. Institutional wording for this boundary statement must be finalized before submission.

Author Contributions

Ruixiang Mei: Conceptualization, methodology, software, formal analysis, investigation, data curation, visualization, writing – original draft, and writing – review and editing. Jianhua Huang: Supervision and writing – review and editing. Final CRediT roles should be confirmed before submission.

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Conflict of Interest

The authors declare no competing interests.

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